Pràctica 1

1. The radiated field by an antenna is given by the following expression:

$$\vec{E} = \frac{e^{-jkr}}{r} E_0 \sin\theta\, \hat{\theta}$$

Find the directivity of the antenna.

$$D(\theta,\phi) = \frac{W(\theta,\phi)}{\frac{P_{rad}}{4\pi r^2}} = 4\pi \frac{U(\theta,\phi)}{P_{rad}}$$

$$D = \frac{4\pi}{\iint E(\theta,\phi)\, d\Omega} = \frac{4\pi}{\int_0^{2\pi}\int_0^{\pi} \sin^3\theta\, d\theta\, d\phi} = \frac{4\pi}{\frac{8}{3}\pi} = \frac{3}{2} \qquad D(dB) = 10\log_{10}(3/2) = 1'7609\, dB$$

$$\int_0^{2\pi}\int_0^{\pi} \sin^3\theta\, d\theta\, d\phi = 2\pi\int_0^{\pi} \sin^3\theta\, d\theta = 2\pi\int_0^{\pi} (1-\cos^2\theta)\sin\theta\, d\theta = 2\pi\left[-\cos\theta + \frac{\cos^3\theta}{3}\right]_0^{\pi} = \frac{8}{3}\pi$$

2. Find the expression of the radiated field by an elemental dipole of length ℓ ($\ell << \lambda$) oriented along the z axis, with a current I_1, placed at the coordinate origin.

$$E_\phi = 0 \qquad E_r = 0$$

$$E_\theta = -jk\eta \frac{e^{-jkr}}{4\pi r} \underbrace{N_\theta}_{-N_z \sin\theta} = jk\eta \frac{e^{-jkr}}{4\pi r} I\ell \sin\theta \qquad (N_z = I\ell)$$

3. Find the expression of the radiated fields by the elemental dipole when it is moved a distance d/2 along the x axis.

$$N = N_0 e^{jk\frac{d}{2}} = I\ell e^{jk\frac{d}{2}}$$

$$\vec{E} = jk\eta \frac{e^{-jkr}}{4\pi r} I\ell \cdot \sin\theta\, e^{jkx\,d/2} = jk\eta \frac{e^{-jkr}}{4\pi r} I\ell \sin\theta\, e^{j\frac{2\pi}{\lambda}\frac{d}{2}} \hat{x}$$

4. Find the expression of the radiated fields by two elemental dipoles oriented along the z axis and displaced along the x axis a distance d (one placed in $x=d/2$ and the other in $x=-d/2$), fed with the same current ($I_1=I_2$). For the distances of d=0.25λ, 0.5 λ and 0.75 λ. Identify the main planes (E-plane and H-plane) and plot them in polar coordinates.

$$N_z = N_1 + N_2 = I\cdot\ell\, e^{-jk\frac{d}{2}}\hat{z} + I\ell\, e^{jk\frac{d}{2}}\hat{z} = I\ell \cdot 2\cos(kd/2)\hat{z}$$

$$N_\theta = -N_z \sin\theta = I2\ell \sin\theta \cos\left(\frac{2\pi}{\lambda}\sin\theta\cos\phi\frac{d}{2}\right)\hat{\theta}$$

$$E_\theta = -jk\eta \frac{e^{-jkr}}{4\pi r} N_\theta = jk\eta \frac{e^{-jkr}}{4\pi r} \sin\theta\, I\cdot 2\ell \cos\left(\frac{\pi}{\lambda}\sin\theta\cos\phi\, d\right)\hat{\theta} = j\frac{I\ell\eta\, e^{-jkr}}{\lambda r}\sin\theta\cos\left(\frac{\pi}{\lambda}\sin\theta\cos\phi\, d\right)\hat{\theta}$$

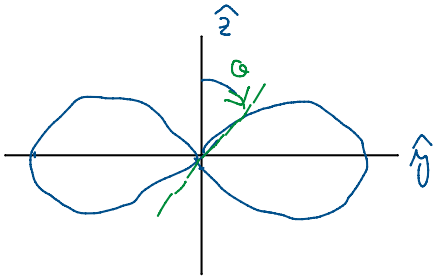
For the different distances $d = 0'25\lambda,\ 0'5\lambda,\ 0'75\lambda$ we identify the main planes

$$f(\theta,\phi) = \sin^2\theta\cos^2\left(\frac{\pi}{\lambda}d\sin\theta\cos\phi\right)\hat{\theta}$$

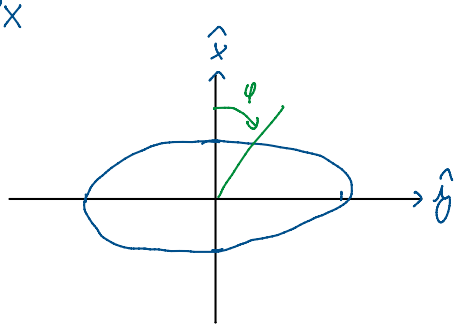
$d = \lambda/4$

$t(\theta,\phi) = \sin^2\theta \cos^2\left(\frac{\pi}{4}\sin\theta\cos\phi\right)$, then we know that maximums are at $\phi = \frac{\pi}{2}$ $\theta = \pi/2 \Rightarrow \hat{y}$

Plane E → YZ



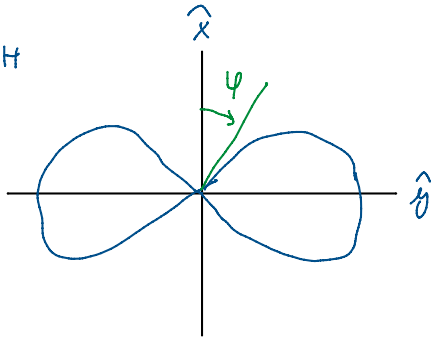
Plane H → YX



$d = \lambda/2$

$t(\theta,\phi) = \sin^2\theta \cos^2\left(\frac{\pi}{2}\sin\theta\cos\phi\right) = \cos^2\left(\frac{\pi}{2}\cos\phi\right)$

Plane H



$d = \lambda/4$ $\quad t(\theta,\phi) = \cos^2\left(\frac{3\pi}{4}\cos\phi\right)$

